

Sarv Pariksha

परीक्षा सबकी, तैयारी पक्की · sarvpariksha.in

User Guide — Visitors & Guests

Everything you can do on sarvpariksha.in **without creating an account** — practise real papers in a genuine exam interface, free.

1. What is SarvPariksha?

SarvPariksha (सर्व परीक्षा — "every exam") is a free exam-practice platform built for two audiences: **students** who want to attempt previous-year papers, official NTA mock tests and AI-generated challenge exams in a real computer-based-test (CBT) interface, and **teachers** who want to launch a shareable, auto-graded exam in minutes instead of hand-building a Google Form.

abhay557

Practice on real previous-year papers

[Start practicing →](#)
[Browse exams](#)

Q 1/3

☒ B Mars

Mars appears red because of iron oxide (rust) on its surface, so it is called the Red Planet.

[View all →](#)

 JEE Mains Physics, Chemistry & Math	75 Qs • 3 hrs
 NEET Biology, Chemistry & Physics	200 Qs • 3.5 hrs
 MHT-CET Physics, Chemistry, Math/Bio	150 Qs • 3 hrs
 Topic Based Focus on specific topics & units	Your topics, on demand
 JEE Advanced Physics, Chemistry & Math	75 Qs • 3 hrs
 GATE CS CS, General Aptitude	65 Qs • 3 hrs
 Coding Tests DSA Logic & Output Prediction	30 Qs • 1 hr

Unlimited mock exams

Full-length tests built on previous-year patterns, free forever. No caps, no trial.

Submit and get your score, section-wise breakdown and solutions in seconds.

AI hints & explanations

A nudge when you're stuck mid-exam. Full step-by-step explanations after submission. AI can be removed from settings as well.

AI paper predictor

Custom papers generated from the latest exam trends so you're always practicing what matters.

"The AI mocks helped me focus on my weak areas. I cracked SSC CGL in my first attempt."

PS **Priya Sharma**
SSC CGL 2023 Qualified

"Topic-wise practice and detailed analytics helped me track progress and improve my speed."

Rahul Verma
IBPS PO Selected

"I practiced 30 minutes daily on SarvPariksha and it made a huge difference in my preparation."

VP **Vikram Patel**
Railway Group D Selected

Everything you need to know about SarvPariksha, mock exams, and AI hints.

Is SarvPariksha really free?

Yes. All mock exams, previous-year question papers, scoring and AI explanations are completely free.
No credit card, no trial period, just start practicing.

Do I need to create an account to practice?

Which exams are covered?

How does the AI hint system work?

Can I turn off AI hints?

Can I practice topic-wise instead of full mocks?

Want to organize an exam?

Teachers (and anyone) can build exams from scratch, from the question bank, or with AI — share them via a link, keep them public or code-protected, and track every student's result with analytics.

Start Organizing →

Practice as a Student

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The SarvPariksha home page

2. What you can do without an account

Feature	Where	Notes
Standard Mock Library	Exams → Standard Mock Library	JEE Mains/Advanced, NEET, GATE, MHT-CET, coding & topic-based papers with instant scoring
Official NTA Mock Tests	Exams → Mock Tests	1,700+ real CBT practice papers across 16 exam bodies (UGC-NET, JEE Main, NEET, DUET, SWAYAM...)
Full CBT exam engine	any exam	Sections, question palette, timer or untimed Practice Mode, negative marking, on-screen numeric keypad
Solutions & explanations	after every submit	Every question shows the correct answer with explanation
Verify any certificate	Certificate Verifier	Enter a SARV-serial or scan the QR printed on a certificate
Take a teacher's shared exam	link <code>/take/TE-XXXXXX</code>	Private exams need the organizer's code; your score goes to the organizer

AI features need a free account. As a guest you cannot use AI Exam Generation, AI Hints inside exams, AI Chat or the Paper Predictor — those unlock the moment you create a free account.

Your data stays yours. As a guest, everything — every paper you open, every answer, every score — is saved **only in this browser's local storage**. Nothing is transmitted to or stored on SarvPariksha's servers. The database only becomes involved when you sign in (to sync history) or use multi-user features like teacher exams. Clearing site data clears guest attempts.

3. Take your first mock exam

1. Open **Exams** in the top navigation. The *Standard Mock Library* lists every bundled paper — search or filter by category.

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AB

Mock Assessment Center

Select custom configurations, trigger dynamic AI-generated question sessions, or practice standard previous papers.

Standard Mock Library

AI Exam Generator

Mock Tests

Teacher Exams

Q

Search mock exams by title or subject...

All Exams

JEE Mains

JEE Advanced

NEET

GATE CS

MHT-CET

Coding Tests

Topic Based

JEE-MAINS 2025 - 29 Jan Shift 2

Physics, Chemistry & Math • 76 Qs • 180 min • +4/-1

View Details →

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Standard Mock Library — search, category chips, one-click paper list

2. Click **View Details** on any paper to see its stats — questions, duration, total marks and marking scheme.

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HomeExamsPricingAI ChatPaper PredictorAttemptsAB

← Back to Exams

JEE-MAINS 2025 - 29 Jan Shift 2

Physics, Chemistry & Math • 2025

JEE MAINS

75

Total Questions

180

Duration (min)

300

Total Marks

3

Subjects

Subjects

Physics25 Qs

Chemistry25 Qs

Mathematics25 Qs

✓

Correct Answer

+4

✗

Incorrect Answer

-1

Start Exam →

About JEE-MAINS 2025 - 29 Jan Shift 2

JEE-MAINS 2025 - 29 Jan Shift 2 is a full-length Physics, Chemistry & Math mock test on SarvPariksha with 75 questions.

Exam detail — stats, subjects and marking before you commit

3. Press **Start Exam**. You'll see an NTA-style instruction page. Read the instructions and tick the declaration checkbox — this mirrors the real exam-day flow.

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JEE-MAINS 2025 - 29 Jan Shift 2

Physics, Chemistry & Math

Category

JEE MAINS

Total Questions

75 Questions

Total Marks

300 Marks

Duration

180 Minutes (3 Hours)

Marking Scheme

+4 for Correct, -1 for Incorrect

Please read the instructions carefully before starting the exam

General Instructions:

1. Total duration of this examination is 180 minutes.

2. The clock will be set at the server. The countdown timer in the top right corner of the screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to manually end or submit your examination.

3. The Questions Palette displayed on the right side of the screen will show the status of each question using one of the following symbols:

1

You have not visited the question yet.

1

You have not answered the question.

1

You have answered the question.

1

You have NOT answered the question, but have marked the question for review.

I have read and understood the instructions. All computer hardware allotted to me are in proper working condition. I declare that I am not in possession of / not wearing / not carrying any prohibited gadget like mobile phone, bluetooth devices etc. / any prohibited material with me into the Examination Hall. I agree that in case of not adhering to the instructions, I shall be liable to be debarred from this Test and/or to disciplinary action.

Practice Mode (No Timer)

Start Timed Simulation

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Pre-start instructions with the mandatory declaration

4. Choose **Start Timed Simulation** (real exam feel) or **Practice Mode** (no timer). Inside the exam: pick answers, use the **question palette** to jump around, **Mark for Review** to flag doubts, and the on-screen keypad for numerical questions.

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JEE-MAINS 2025 - 29 Jan Shift 2

02:59:58

Sections: Physics Chemistry Mathematics

Section A (MCQs)

Section B (Numerical)

Section A (MCQ) — Question No. 1

+4 / -1

AI Hint

0/100

Request Hint

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities . If we wait sufficiently long for elastic collision to happen, $v_A = 5m/s$, $v_B = 2m/s$, $v_C = 4m/s$ then will be the final velocities. Reason (R): In an elastic collision $v_A = 4m/s$, $v_B = 2m/s$, $v_C = 5m/s$ between identical masses, two objects exchange their velocities. In the light of the above statements, choose the correct answer from the options given below :

A

B

C

$v_A = 5$

$v_B = 2$

$v_C = 4$

☐ (A) (A) is false but (R) is true

☐ (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

☐ (C) Both (A) and (R) are true and (R) is the correct explanation of (A)

☐ (D) (A) is true but (R) is false

Mark for Review & Next

Clear Response

Back

Save & Next

A

abhay mourya

Candidate

YOU ARE VIEWING PHYSICS SECTION

QUESTION PALETTE

SECTION A (MCQS)

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

SECTION B (NUMERICAL)

1

2

3

4

5

6

7

8

9

Answered

Not Answered

Marked

Marked & Answered

Not Visited

Submit Test

The CBT engine — sections, palette, timer, AI hint panel

5. Press **Submit Test** and confirm. Your scorecard appears instantly with correct/incorrect/skipped breakdown, followed by full solutions for every question.

SarvPariksha

JEE-MAINS 2025 - 29 Jan Shift 2

02:59:56

Sections: Physics Chemistry Mathematics

Section A (MCQs)

Section B (Numerical)

Section A (MCQ) — Question No. 1

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities . If we wait sufficiently long for elastic collision to happen, $v_A = 5m/s$, $v_B = 2m/s$, $v_C = 4m/s$ then will be the final velocities. Reason (R): In an elastic collision $v_A = 4m/s$, $v_B = 2m/s$, $v_C = 5m/s$ between identical masses, two objects exchange their velocities. In the light of the above statements, choose the correct answer from the options given below :

A

B

C

$v_A = 5$

$v_B = 2$

$v_C = 4$

☐ (A) (A) is false but (R) is true

☒ (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

☐ (C) Both (A) and (R) are true and (R) is the correct explanation of (A)

☐ (D) (A) is true but (R) is false

Mark for Review & Next

Clear Response

Back

Save & Next

A

abhay mourya

Candidate

YOU ARE VIEWING PHYSICS SECTION

QUESTION PALETTE

SECTION A (MCQS)

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

SECTION B (NUMERICAL)

1

2

3

4

5

6

7

8

9

Answered

Not Answered

Marked

Marked & Answered

Not Visited

Submit Test

Exam Submission Summary

Please review your attempt details before final submission.

Section Name	No. of Questions	Answered	Not Answered	Marked for Review	Marked & Answered	Not Visited
Physics — Section A (MCQs)	2	1	0	0	0	1
Physics — Section B (Numerical)	23	0	0	0	0	23
Chemistry — Section A (MCQs)	2	0	0	0	0	2
Chemistry — Section B (Numerical)	23	0	0	0	0	23
Mathematics — Section A (MCQs)	2	0	0	0	0	2
Mathematics — Section B (Numerical)	23	0	0	0	0	23

Once submitted, you will not be able to modify your answers or resume the test.

Close & Resume Test

Submit Exam

Submit confirmation — shows an answered/unanswered summary

Performance Scorecard

JEE-MAINS 2025 - 29 Jan Shift 2



Correct	0
Incorrect	1
Skipped	74
Time Taken	0m 4s
Total Marks	300



Standard Practice Mock Exam

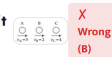
This is a standard free practice mock exam. Official cryptographically verifiable accomplishment certificates are only awarded for custom AI-Generated exams scoring $\geq 40\%$.

Solutions & Explanations

Physics — Section A — Q1.

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities . If we wait sufficiently long for elastic collision to happen, $vA = 5m/s$, $vB = 2m/s$, $vC = 4m/s$ then will be the final velocities.

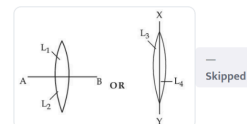
Reason (R): In an elastic collision $vA = 4m/s$, $vB = 2m/s$, $vC = 5m/s$ between identical masses, two objects exchange their velocities. In the light of the above statements, choose the correct answer from the options given below :



Correct Answer: A

Physics — Section A — Q2.

Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is



Correct Answer: B

Physics — Section B — Q1.

Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2 , respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is



Correct Answer: B

Physics — Section B — Q2.

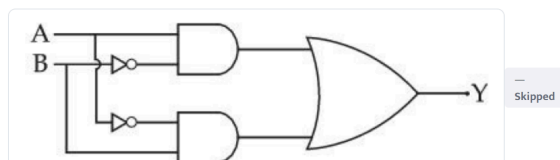
Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process. Reason (R) : In isothermal process, constant, $PV = PV^\gamma$ while in adiabatic process constant. Here is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas. In the light of the above statements, choose the correct answer from the options given below :



Correct Answer: A

Physics — Section B — Q3.

The truth table for the circuit given below is :



Correct Answer: D

Physics — Section B — Q4.

The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have



Correct Answer: D

Physics — Section B — Q5.

Match List - I with List - II. Choose the correct answer from the options given below :

List - I	List - II
(A) Young's Modulus	(I) $\text{M L}^{-1} \text{T}^{-1}$
(B) Torque	(II) $\text{M L}^{-1} \text{T}^{-2}$
(C) Coefficient of Viscosity	(III) $\text{M}^{-1} \text{L}^3 \text{T}^{-2}$
(D) Gravitational Constant	(IV) $\text{M L}^2 \text{T}^{-2}$

— Skipped

Correct Answer: C

Physics — Section B — Q6.

A sand dropper drops sand of mass $m(t)$ on a conveyor belt at a rate proportional to the square root of speed (v) of the belt, i.e. $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

— Skipped

Correct Answer: C

Physics — Section B — Q7.

Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in a free space. At $t = 0$, they are given an initial velocity $\vec{V}_A = V_0 \vec{AC}$, $\vec{V}_B = V_0 \vec{BA}$ and $\vec{V}_C = V_0 \vec{CB}$. Here, $\vec{AC}$, $\vec{CB}$ and $\vec{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is:



— Skipped

Correct Answer: C

Physics — Section B — Q8. The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level, is

— Skipped

Correct Answer: C

Physics — Section B — Q9.

A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

— Skipped

Correct Answer: B

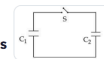
Physics — Section B — Q10. In an experiment with photoelectric effect, the stopping potential,

— Skipped

Correct Answer: D

Physics — Section B — Q11.

A capacitor, $C_1 = 6\mu\text{F}$ is charged to a potential difference of $V_0 = 5\text{V}$ using a 5V battery. The battery is removed and another capacitor, $C_2 = 12\mu\text{F}$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the



— Skipped

Correct Answer: A

Physics — Section B — Q12.

A plane electromagnetic wave propagates along the +x direction in free space. The components of the electric field, $\vec{E}$ and magnetic field, $\vec{B}$ vectors associated with the wave in Cartesian frame are

— Skipped

Correct Answer: B

Physics — Section B — Q13.

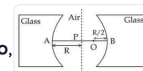
A cup of coffee cools from 90° C to 80° C in t minutes when the room temperature is 20° C. The time taken by the similar cup of coffee to cool from 80° C to 60° C at the same room temperature is :

— Skipped

Correct Answer: D

Physics — Section B — Q14.

Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :

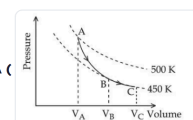


— Skipped

Correct Answer: D

Physics — Section B — Q15.

A poly-atomic molecule ($C_V = 3R$, $C_P = 4R$, where R is gas constant) goes from phase space point A ($P_A = 10^5 \text{ Pa}$, $V_A = 4 \times 10^{-6} \text{ m}^3$) to point B ($P_B = 5 \times 10^4 \text{ Pa}$, $V_B = 6 \times 10^{-6} \text{ m}^3$) to point



— Skipped

Correct Answer: B

Physics — Section B — Q16.

Match List - I with List - II. List - I List - II (A) Magnetic induction (I) Ampere meter (B) Magnetic intensity (II) Weber (C) Magnetic flux (III) Gauss (D) Magnetic moment (IV) Ampere/meter

— Skipped

Correct Answer: D

Physics — Section B — Q17.

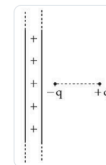
A point charge causes an electric flux of $-2 \times 10^4 \text{ Nm}^2 \text{ C}^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is: (Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

— Skipped

Correct Answer: D

Physics — Section B — Q18.

An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0 . Choose the correct option from the following.



— Skipped

Correct Answer: D

Physics — Section B — Q19.

The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is _____ π cm.

— Skipped

Correct Answer: 8

Physics — Section B — Q20.

A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is (Take, $\epsilon_0 = 9 \times 10^{-12} \frac{\text{F}}{\text{m}}$, $\pi = \frac{22}{7}$) _____ μm .

— Skipped

Correct Answer: 1320

Physics — Section B — Q21.

Two planets, A and B are orbiting a common star in circular orbits of radii R_A and R_B , respectively, with $R_B = 2R_A$. The planet B is $4\sqrt{2}$ times more massive than planet A. The ratio $\left(\frac{L_B}{L_A}\right)$ of angular momentum (L_B) of planet B to that of planet A (L_A) is closest to integer _____.

— Skipped

Correct Answer: 8

Physics — Section B — Q22.

Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time $t = 0$, for the first time. The maximum possible number of crossing(s) (including the crossing at $t = 0$) is _____.

— Skipped

Correct Answer: 3

Physics — Section B — Q23.

A physical quantity Q is related to four observables a, b, c, d as follows: $Q = \frac{ab^4}{cd}$ where, $a = (60 \pm 3)\text{Pa}$; $b = (20 \pm 0.1)\text{m}$; $c = (40 \pm 0.2)\text{Nsm}^{-2}$ and $d = (50 \pm 0.1)\text{m}$, then the percentage error in Q is $\frac{x}{1000}$, where $x =$ _____.

— Skipped

Correct Answer: 7700

Chemistry — Section A — Q1.

If the pressure applied over the system $\text{CO}(g) + 3\text{H}_2(g) \rightleftharpoons \text{CH}_4(g) + \text{H}_2\text{O}(g)$ increases by two fold at constant temperature then (A) Concentration of reactants and products increases. (B) Equilibrium will shift in forward direction. (C) Equilibrium constant increases since concentration of products increases. (D) Equilibrium constant remains unchanged as concentration of reactants and products remain same. Choose the correct answer from the options given below :

— Skipped

Correct Answer: B

Chemistry — Section A — Q2.

Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16mg/mL which becomes 4mg/mL in 12 months. The expiry time of the drug in months is _____. Assume that the decomposition of the drug follows first order kinetics.

— Skipped

Correct Answer: B

Chemistry — Section B — Q1. O_2 gas will be evolved as a product of electrolysis of:

— Skipped

Correct Answer: B

Chemistry — Section B — Q2. Identify the homoleptic complexes with odd number of d electrons in the central metal :

— Skipped

Correct Answer: C

Chemistry — Section B — Q3. Which one of the following, with HBr will give a phenol?

— Skipped

Correct Answer: C

Chemistry — Section B — Q4.

Given below are two statements: Statement (I): NaCl is added to the ice at 0°C , present in the ice cream box to prevent the melting of ice cream. Statement (II): On addition of NaCl to ice at 0°C , there is a depression in freezing point. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: D

Chemistry — Section B — Q5.

Match List - I with List - II : Choose the correct answer from the options given below :

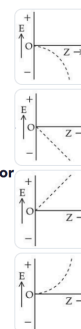
List - I	List - II
Applications	Batteries/Cell
(A) Transistors	(i) Anode - Zn/Hg ; Cathode - $\text{HgO} + \text{C}$
(B) Hearing aids	(ii) Hydrogen fuel cell
(C) Invertors	(iii) Anode - Zn ; Cathode - Carbon
(D) Apollo space ship	(iv) Anode - Pb ; Cathode - Pb/ PbO_2

— Skipped

Correct Answer: C

Chemistry — Section B — Q6.

For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n ? [E: Energy of the stationary state, Z : atomic number, n = principal quantum number]



— Skipped

Correct Answer: A

Chemistry — Section B — Q7. Which one of the following reaction sequences will give an azo dye?

— Skipped

Correct Answer: A

Chemistry — Section B — Q8. Identify the essential amino acids from below :

— Skipped

Correct Answer: B

Chemistry — Section B — Q9. The calculated spin-only magnetic moments of $\text{K}_3[\text{Fe}(\text{OH})_6]$ and $\text{K}_4[\text{Fe}(\text{OH})_6]$ respectively are :

— Skipped

Correct Answer: D

Chemistry — Section B — Q10.

The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

— Skipped

Correct Answer: B

Chemistry — Section B — Q11. Total number of sigma (σ) and pi (π) bonds respectively present in hex-1-en-4-yne are :

— Skipped

Correct Answer: C

Chemistry — Section B — Q12.

First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

— Skipped

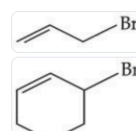
Correct Answer: C

Chemistry — Section B — Q13.

Given below are two statements: Statement (I): On nitration of m-xylene with HNO_3 , H_2SO_4 followed by oxidation, 4-nitrobenzene-1,3-dicarboxylic acid is obtained as the major product. Statement (II): $-\text{CH}_3$ group is o/p-directing while $-\text{NO}_2$ group is m-directing group. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

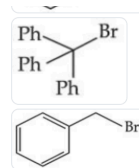
Correct Answer: D



Chemistry — Section B — Q14.

Chemistry — Section B — Q14.

Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?



— Skipped

Correct Answer: C

Chemistry — Section B — Q15.

Given below are two statements: Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle. Statement (II): If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geq \sqrt{\frac{h}{\pi} \times \frac{1}{2m}}$. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: C

Chemistry — Section B — Q16.

0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation $5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$ Identify the correct statements:

— Skipped

Correct Answer: A

Chemistry — Section B — Q17.

Given below are two statements: Statement (I): In partition chromatography, stationary phase is thin film of liquid present in the inert support. Statement (II): In paper chromatography, the material of paper acts as a stationary phase. In the light of the above statements, choose the correct answer from the options given below:

— Skipped

Correct Answer: A

Chemistry — Section B — Q18.

If C (diamond) $\rightarrow C$ (graphite) + $XkJmol^{-1}$ C (diamond) + O_2 (g) $\rightarrow CO_2$ (g) + $YkJmol^{-1}$ C (graphite) + O_2 (g) $\rightarrow CO_2$ (g) + $ZkJmol^{-1}$ at constant temperature. Then

— Skipped

Correct Answer: D

Chemistry — Section B — Q19.

In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is ____ $\times 10^{-1}\%$. (Molar mass : O = 16, S = 32, Ba = 137 in $gmol^{-1}$)

— Skipped

Correct Answer: 275

Chemistry — Section B — Q20.

Isomeric hydrocarbons $\rightarrow$ negative Baeyer's test (Molecular formula C_9H_{12}) The total number of isomers from above with four different non-aliphatic substitution sites is -

— Skipped

Correct Answer: 2

Chemistry — Section B — Q21.

Consider the following low-spin complexes $K_3[Co(NO_2)_6]$, $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $Cu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$. The sum of the spin-only magnetic moment values of complexes having yellow colour is. ____ B.M. (answer in nearest integer)

— Skipped

Correct Answer: 0

Chemistry — Section B — Q22.

In the Claisen-Schmidt reaction to prepare, dibenzalacetone from 5.3 g of benzaldehyde, a total of 3.51 g of product was obtained. The percentage yield in this reaction was ____%.

— Skipped

Correct Answer: 60

Chemistry — Section B — Q23. Total number of non bonded electrons present in ion based on Lewis theory is

— Skipped

Correct Answer: 12

Mathematics — Section A — Q1. Let $f(x) = \int_0^x t(t^2 - 9t + 20)dt$, $1 \leq x \leq 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals:

— Skipped

Correct Answer: D

Mathematics — Section A — Q2.

Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$, and makes an angle of $\cos^{-1}(-\frac{1}{3})$ with the vector $\hat{i} + \hat{j} + \hat{k}$. If $\hat{a}$ makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i} + \alpha\hat{j} + \hat{k}$, then the value of α is:

— Skipped

Correct Answer: B

Mathematics — Section B — Q1.

If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2 + \sec x}{(1 + 2 \sec x)^2}$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $f(\frac{\pi}{3}) = \frac{\sqrt{3}}{10}$, then $f(\frac{\pi}{4})$ is equal to:

— Skipped

Correct Answer: D

Mathematics — Section B — Q2.

Let P be the foot of the perpendicular from the point $(1, 2, 2)$ on the line $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the line $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$, $\lambda \in \mathbb{R}$, intersect the line L at Q . Then $2(PQ)^2$ is equal to :

—
Skipped

Correct Answer: D

Mathematics — Section B — Q3.

Let $A = [a_{ij}]$ be a matrix of order 3×3 , with $a_{ij} = (\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}$, $\alpha, \beta \in \mathbb{Z}$, then $\alpha + \beta$ is equal to:

—
Skipped

Correct Answer: B

Mathematics — Section B — Q4.

Let the line $x + y = 1$ meet the axes of x and y at A and B , respectively. A right angled triangle AMN is inscribed in the triangle OAB , where O is the origin and the points M and N lie on the lines OB and AB , respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and $AN : NB = \lambda : 1$, then the sum of all possible value(s) of λ :

—
Skipped

Correct Answer: A

Mathematics — Section B — Q5.

If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440^{th} position in this arrangement, is :

—
Skipped

Correct Answer: C

Mathematics — Section B — Q6.

If the set of all $a \in \mathbb{R}$, for which the equation $2x^2 + (a-5)x + 15 = 3a$ has no real root, is the interval (α, β) , and $X = \{x \in \mathbb{Z} : \alpha < x < \beta\}$, then $\sum_{x \in X} x^2$ is equal to:

—
Skipped

Correct Answer: D

Mathematics — Section B — Q7.

Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j . Let the random variable X denote the possible values of the determinant of the matrix A . Then, the variance of X is :

—
Skipped

Correct Answer: C

Mathematics — Section B — Q8.

Let the function $f(x) = (x^2 + 1)|x^2 - ax + 2| + \cos|x|$ be not differentiable at the two points $x = \alpha = 2$ and $x = \beta$. Then the distance of the point (α, β) from the line $12x + 5y + 10 = 0$ is equal to :

—
Skipped

Correct Answer: C

Mathematics — Section B — Q9.

Let the area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α . If $9\alpha = \beta\pi + \gamma$; β, γ are integers, then the value of $|\beta - \gamma|$ equals.

—
Skipped

Correct Answer: B

Mathematics — Section B — Q10. The remainder, when 7^{103} is divided by 23, is equal to :

— Skipped

Correct Answer: D

Mathematics — Section B — Q11.

If $\alpha x + \beta y = 109$ is the equation of the chord of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, whose mid point is $(\frac{5}{2}, \frac{1}{2})$, then $\alpha + \beta$ is equal to:

—
Skipped

Correct Answer: A

Mathematics — Section B — Q12.

If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ, δ) , then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

—
Skipped

Correct Answer: C

Mathematics — Section B — Q13.

Let a circle C pass through the points $(4, 2)$ and $(0, 2)$, and its centre lie on $3x + 2y + 2 = 0$. Then the length of the chord, of the circle C , whose mid-point is $(1, 2)$, is:

—
Skipped

Correct Answer: C

Mathematics — Section B — Q14.

Let a straight line L pass through the point $P(2, -1, 3)$ and be perpendicular to the lines $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ and $\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$. If the line L intersects the yz -plane at the point Q , then the distance between the points P and Q is:

—
Skipped

Correct Answer: D

Mathematics — Section B — Q15.

Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains n white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is $29/45$, then n is equal to :

—
Skipped

Correct Answer: A

Mathematics — Section B — Q16.

Let $\alpha, \beta (\alpha \neq \beta)$ be the values of m , for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10} (n^\alpha + n^\beta)$ is equal to:

—
Skipped

Correct Answer: D

Mathematics — Section B — Q17.

Let $S = \mathbb{N} \cup \{0\}$. Define a relation R from S to $\mathbb{R}$ by: $\mathbf{R} = \left\{ (x, y) : \log_e y = x \log_e \left(\frac{2}{5} \right), x \in S, y \in \mathbb{R} \right\}$. Then, the sum of all the elements in the range of R is equal to:

—
Skipped

Correct Answer: D

Mathematics — Section B — Q18.

If $\sin x + \sin^2 x = 1, x \in (0, \frac{\pi}{2})$, then $(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ is equal to :

—
Skipped

Correct Answer: D

Mathematics — Section B — Q19.

If $24 \int_0^{\frac{\pi}{12}} (\sin |4x - \frac{\pi}{12}| + [2 \sin x]) dx = 2\pi + \alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to _____.

—
Skipped

Correct Answer: 12

Mathematics — Section B — Q20.

Let $a_1, a_2, \dots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \dots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \dots + a_{2024}$ is equal to

—
Skipped

Correct Answer: 11132

Mathematics — Section B — Q21. If $\lim_{t \rightarrow 0} \left(\int_0^1 (3x + 5)^t dx \right)^{\frac{1}{t}} = \frac{a}{5e} \left(\frac{8}{5} \right)^{\frac{2}{3}}$, then α is equal to

— Skipped

Correct Answer: 64

Mathematics — Section B — Q22.

Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ) = \frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to _____.

—
Skipped

Correct Answer: 1328

Mathematics — Section B — Q23.

Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b) , for which $\left| \frac{z-a}{z+b} \right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in \mathbb{C}$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to _____.

—
Skipped

Correct Answer: 10



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